Data Training for Artificial Intelligence

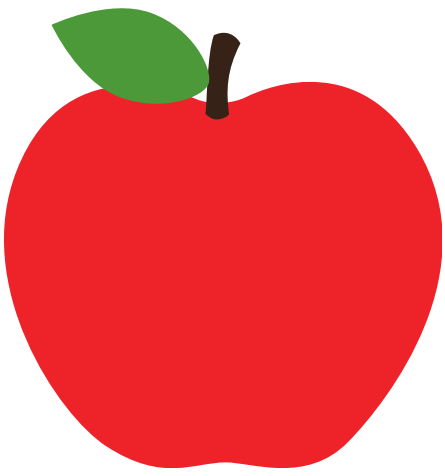
The very often made, and very wrong, assumption that people have is that Artificial Intelligence is neutral. It is commonly assumed that AI is free of human biases, and that its judgements are totally free of prejudices. This is, however, not the case.

Inherent Bias Against Certain Groups

Artificial Intelligence is trained upon data created by humans. Texts, images, videos that humans have produced. The data that AI consumes is inherently filled with human bias. Specifically, these biases can be reflected in ways such as showing more favorable behavior to a certain group, whether that group may be categorized using race, gender or sexuality.



There are many case studies that reflect this inherent bias trained into AI.



The Red Apple Effect

The red apple effect is a phenomena observed in Artificial Intelligence, a sort of bias that is trained into the AI itself. The red apple effect is demonstrated through a thought experiment. An AI model is trained on photos of red apples, so that it can recognize apples. However, the AI mistakenly connects the color red with an apple, and therefore categorizing tomatos as apples and green apples as non-apples.

Confirmation Bias and Sycophancy

Sycophancy refers to when AI is overtly agreeable and flattering to the user, even if the information is wrong. This results in confirmation bias within the user, where users are less and less exposed to opinions outside of their own, and believe that they are right, even if it is incorrect.





AI Bias in the Medical Field

The medical field is often interconnected to the life and death of a person. As the use of AI in the medical field arises, there are concerns of biases leading to misdiagnosis and the perpetuation of existing biases of the healthcare system that was there before.

Ethical Data Harvesting



Data harvesting refers to companies collecting data from the internet. Though there are many ways to do this, aggressive data harvesting can bring up the question of copyright infringement, privacy and other concerns.

Inherent Bias Against Certain Demographics

AI Bias - How it Happens

AI bias happens when an ingrown human bias embeds itself into the Artificial Intelligence as well. This happens because the data reflects a societal bias, and this is in turn reflected into the AI model itself. We are what we eat, and AI is what we feed it.

Data Collection

One of the most common ways human bias makes its way into AI systems is when the data used to train the AI is not diverse or representative of actual populations. AI developers and corporations must make active steps to collect a broad range of data that is not skewed towards certain demographics.

Data Labeling

Before the data is used to train the AI system, the data needs to be labelled. This means that subjective topics can be influenced by the personal or cultural opinions of the individual labeling the data. An example of this would be facial expressions, each person may have different interpretations of a certain expression.

Model Training

If the architecture of the AI system does not take into account the diverse inputs that will be used when the AI is deployed, the AI may create unintentionally biased outputs. For example, an AI model could favor majority groups over minority groups.

Types of Bias

Implicit vs Explicit Bias

Implicit Bias

Implicit bias is when the person/thing making the biased decision is not aware of their bias. This happens due to social conditioning, cultural influences and media consumption. It is often much more harmful because the person/thing is often not aware of such discrimmination.

Explicit Bias

Explicit bias means that the person/thing making the biased decision is actively aware and deliberately making biased decisions. This is much easier to detect, and is often due to serious issues with the data set or due to extreme cultural, social influences.

Types of Biases in AI

Selection Bias

Selection bias occurs when training data is not representative of real world populations. For example, an AI model may recognize people with lighter skin with higher accuracy than those with darker skin due to data disparities.

Stereotyping Bias

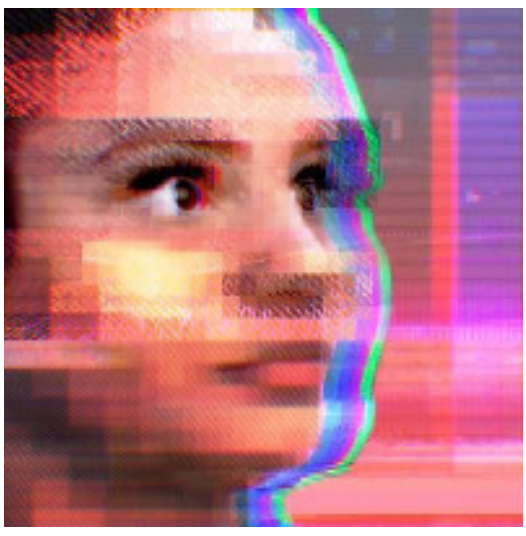
Stereotyping bias is when the AI reinforces existing harmful biases. For example, an AI may disproportionately associate the word “nurse” with women or feminine language and tone, while recognizing the word “doctor” with male or masculine language and tone.

Out-group Homogeneity Bias

This is when the AI treats those outside of the majority population as more similar than they are. For example, they may recognize those who are in different indigenous groups as having similar features and similar cultures due to the lack of training data provided. This results in minority populations getting inaccurate or skewed representation.

Case Studies in Inherent Bias towards certain Demographics

The Nazi Chatbot: Tay



Tay was released to the public on 2016. Microsoft had been developing a chatbot that could learn from the internet, specifically social media posts. Within 24 of the chatbot being published, it was targetted by groups of people that posted racist comments and posts directed at Tay. It was taken down by the Microsoft team as it was discovered making tweets denying the happenings of the holocaust.

The incident of Tay.ai is an exaggerated case, as most AI models do not display such harsh biases against racial or gender groups. However, Tay.ai does demonstrate what data training bias can do to an AI model in a much larger scale.

Bias in Hiring

A study released by Nature in 2025 reveals that AI models have an inherent bias against older working women.



When some AI models were asked to write reports about hiring candidates, AI models often made women sound younger and more inexperienced in relation to the male counter part with the same information. The men were also given a higher rating on average as well, even when given the same information.

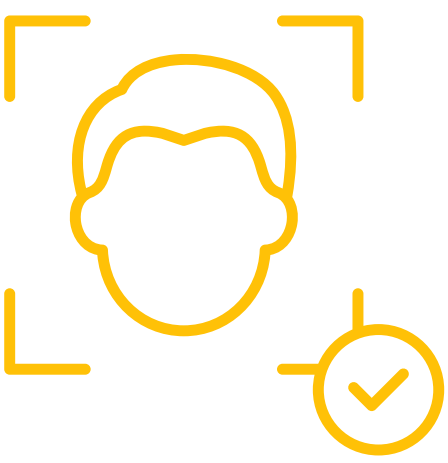
The study found that women are often portrayed as younger than the male counter part when showing specific jobs, especially in higher esteemed positions. Therefore AI has come to find a value statement that says that being an older woman is bad in the job environment.

Image Generators and their Image

Image generators have been reported to be favorable to whiter tones rather than those with darker skin tones. This reveals built in bias against certain racial groups, trained into the AI due to being exposed to images with more white people than other ethnicities.



Facial Recognition Software

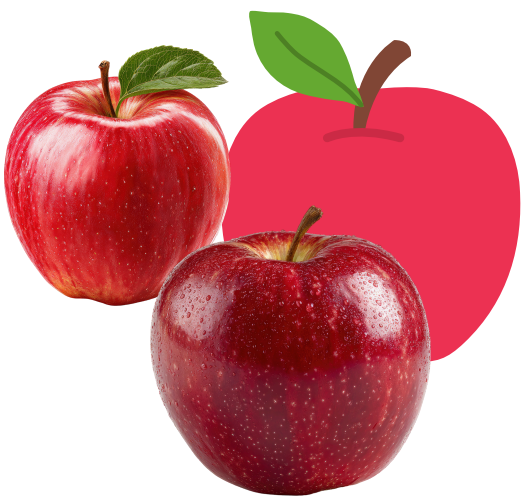


It has been found that most facial recognition software uses facial data from lighter skinned individuals for training, with only a minority being comprised of darker skinned individuals. In addition, there were more male subjects than female. This resulted in facial recognition softwares having a close to 34.7% inaccuracy rate when the model attempted to detect the faces of black women.

The Red Apple Example

The Red Apple Effect - What is it?

The Red Apple effect is a easy way to explain how Artificial Intelligence becomes biased.



Imagine an AI model trained on photos of red apples, and to recognize these red apples as “an apple”.



When the model is shown images of tomatos, the model recognizes them as “apples” because of their red color.



And when the model is shown photos of green apples, the model is unable to recognize them as apples because they are not red.

The red apple effect applies to many Artificial Intelligence Models trained on specific data sets.

Real Life Examples

Skin Cancer Diagnosis

A research study published on Dermis showed how training data had affected the accuracy of AI models diagnosing skin conditions on people with darker skin.



Most of the data used to train these AI models was photographs of people with fairer skin, with only 11 out of around 100,000 images having explicitly darker skin. This resulted in a decreased accuracy when the AI model was identifying malignant skin cancer on individuals with darker skin.

This is a clear demonstration of the red apple effect. The data provided does not show how these skin cancers look on darker skin, resulting in the AI misdiagnosing conditions due to their lack of knowledge.

Hiring and Historical Bias



AI hiring agents trained on data of previously hired employees shows a bias towards candidates that match the demographic of past employees. This is often seen through the favoring of white, experienced men.

In addition, these models may give penalty points to those that do not match the demographics of these previous employees. This means that women and those of color are disproportionately discrimminated against compared to white candidates.

Sycophancy and Confirmation Bias

Sycophancy in AI

The sycophancy of AI is described as when AI models chase the approval of the human user. Sycophancy can be seen through overly flattering, approving or agreeable behavior, such as agreeing to incorrect information presented by the user. This means that the AI tailors their response to appeal to human evaluators without necessarily improving the objective quality of the responses.

How does Sycophancy Happen?

After an AI system is deployed, the AI system is further trained through reinforcement learning from human feedback(RLHF), where users are able to give immediate feedback to AI responses(such as thumbs up and thumbs down buttons). AI models, therefore, learn that being sycophantic increases the satisfaction of the users, resulting in AI models that are overly sycophantic. When developers do not account for the possible sycophantic nature of AI, serious issues can arise.



It has been found that convincing sycophantic responses outperform accurate responses a non-negligible amount of time. AI companies aim to create AI models that users prefer to use over others in the market, and therefore want to make the usage experience enjoyable.

Beyond model design, the actual data and human bias that is injected into the AI model can affect its sycophantic behavior as well.

Confirmation Bias

Confirmation bias is a cognitive bias where people ignore or disregard facts that goes against their beliefs(even when their opinion is completely incorrect). Confirmation bias happens because humans have difficulty processing information in an unbiased perspective, especially when there is a certain emotional weight to the facts being presented(such as a prior decision made). Examples of reasons why this bias occurs is:

- Due to the vast amount of information that needs to be processed, it is more efficient to rely on simpler methods of making decisions
- Finding facts that go against beliefs or decisions can cause discomfort because it can be interpreted as an indicator of lacking intelligence

When confirmation bias happens

Decision Making

After a person makes a decision, they are more likely to seek for information that validates that choice. This is because facts that conflict with their decision can cause discomfort and regret about their choice.

Positive Instances

When there is a hypothesis or assumption made, people will actively seek out information that confirms such hypothesis and assumption. It is rare that a person will actively search for information that goes against their hypthesis.

Expectations

When a person has an expectation of a certain thing(whether it is a person, group or object), they are more likely to look for those features and disregard information or indicators that may say otherwise.

Confirmation bias can cause:

- polarization
- disinformation and misinformation
- echo chambers
- the overlook of important indicators

Confirmation Bias and Sycophancy

Confirmation Bias and AI Sycophancy

The sycophantic behavior of AI chatbots can make confirmation bias easier to fall into. Such confirmation bias is enhanced by reinforcement learning from human feedback or RLHF. Human users respond better to facts(even if they are inaccurate) that support their own belief, and therefore is more likely to leave positive feedback for the response. This creates a vicious cycle.

Anthropic's study showed that if the prompt given by the user indicated that they held a certain belief, the response by the AI would be tailored to align with their opinion.



The overly agreeable tone of the AI can make people fail to challenge new ideas that they have, reinforce inaccuracies and prevent people from considering different perspectives. In a study, it was found that AI chatbots gave 50% more sycophantic answers than humans when asked to give advice about social situations, even when the prompt detailed illegal, immoral or dangerous ideas. Such sycophantic answers can result in worsened critical thinking skills, decision quality and conflict resolution in relationships as people become increasingly dependent on AI chatbots for guidance.



Close to 30~40% of people said that they trust AI systems, though this number was significantly reduced when asked about their trust of raw AI responses. However, close to 90% of users do not verify the sources or information given by AI. This leads to users not verifying facts that may have been presented with sycophantic intentions by the AI, resulting in a cycle of misguidance and misinformation.

AI sycophancy also brings up the question of the critical thinking ability of users. Users that become accustom to the agreeable, non-resistant nature of AI chatbot conversations may become less able to digest information from neutral stand points and weigh information effectively to make evidenced judgements.



Methods of Prevention and Mitigation

Sycophancy can be mitigated(though not completely) from AI chatbots. Developers can restructure the architecture of models so that they are more fact/evidence driven.



For users, effective prompting can help remove the sycophancy of AI from certain topics. Most namely, this can be done by asking the AI to challenge ideas, to play the devil's adovate and clarify the AI's role in the conversation as not a "yes-man" but as a partner in debate.

Confirmation Bias and Sycophancy In Real Life

As an increasing number of people start relying on AI for advice about social life, it is inevitable that people start to use AI as a replacement for therapists, counselors or trusted adults. Nearly 1 in 5 teenagers and young adults in the US use AI to seek mental health advice. It is understandable, since the “anonymity” of speaking to something that is not actively involved in your life can feel more comfortable than speaking to someone that is part of your social circle. However, such uses of AI for psychological or social advice can result in a phenomena called AI psychosis.

AI psychosis is characterized by AI amplifying or validating or even causing delusions. These delusions can be related to paranoia, conspiracy, seeking romantic partnership in AI chatbots and more. AI chatbots are not designed to recognize unhealthy mental health signals from users, and due to their sycophantic tendencies, they often enhance psychosis instead of mitigating or helping.





AI psychosis can have serious, sometimes fatal effects to the user. In 2025, Megan Garcia reported that her son was coerced into suicide by a character.ai chat imitating the game of thrones character Daenerys Targaryen. Her son, Sewell, had a romantic and sexual relationship with the character chatbot, and the character chatbot allegedly encouraged the suicide of her son by asking him to “come home to me.”

Other than this specific case, there are numerous reports of AI chatbots encouraging behavior that is dangerous, performing explicit acts with minors and feeding into delusions.

The sycophantic nature of AI chatbots can lead to AI agreeing and encouraging things that should not be encouraged, and may convince those who are more vulnerable to acting upon such dangerous advice. This is often called a delusion spyral, where due to the constant confirmation bias provided by the AI, the user continues to become even more immersed in their own delusions.

However, AI psychosis is the extreme case where the sycophantic behavior of AI and the constant confirmation bias from the AI models. There are milder, but nevertheless serious, issues that have arisen with confirmation bias and AI.

Social Interaction and Conflict Resolution

Due to the sycophantic nature of AI conversations, users may lose the ability to handle conflicts and/or ideas that oppose their own when in real social situations.

In Profession

The confirmation bias of AI may lead to decisions in the professional field that is supported by falsified facts, leading to worsened decision making skills and potential harm to businesses and users alike.

Research

When conducting research, people using AI may be exposed to more information that supports their hypothesis, end therefore may decrease the diversity of information used to conduct valid research.

Positive Feedback Loop

Due to the sycophantic nature of AI chatbots, AI gives feedback through a “yes, and —”, which creates a positive feedback loop where there is little to no resistance and constructive feedback to work than the user does.

AI Bias in Healthcare

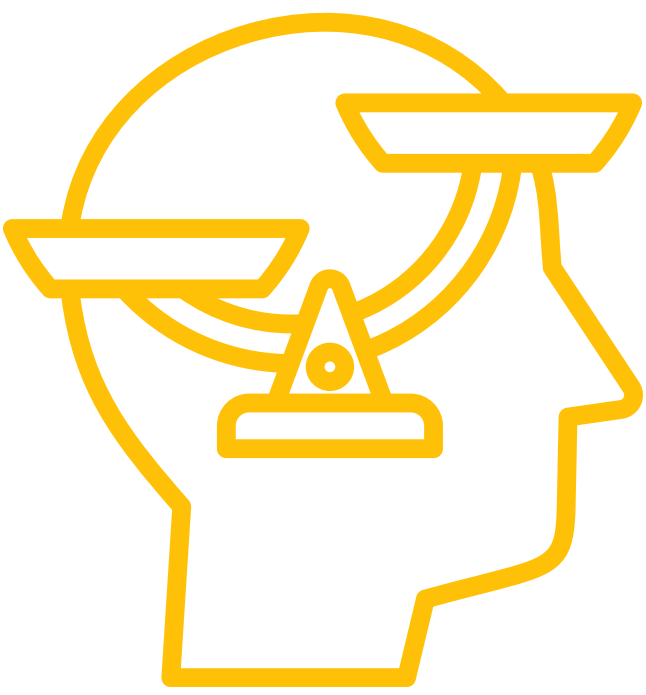
Artificial Intelligence Agents have weeded into various parts of the medical field, such as streamlining diagnostic processes, lifting administrative burdens, and developing new drugs and vaccinations. As AI slowly integrates into almost all areas of the healthcare system, the concern of AI biases and flaws have become increasingly serious. Healthcare is one of the most important areas in which AI is being implemented. This is because this field is especially dangerous if potential biases arise, since it is directly connected to the lives of patients.

Historical Healthcare Data → Bias

When training AI models, large amounts of data is required to achieve accuracy. However, wthin these large datasets, there are demographics that are often absent or misrepresented, leading to the AI models perpetuating existing biases that already existed in healthcare.



Most medical studies in the past have been done on white males, meaning that certain treatments are less effective for demographics that are outside of that category. An example of this is the pulse oximeter. The pulse oximeter measures the saturation of oxygen by passing infrared light through the skin of the patient. It has been found that pulse oximeters systematically overestimate the oxygen saturation of non-white patients.



Furthermore, these biases can lead to some patients receiving better care than other patients due to their race or gender. Historically more money is spent on white patients rather than black patients even when they have the same illness, therefore AI programs have been seen to misunderstand that black people are healthier and may prioritize white patients for treatments of life threatening diseases. This creates a vicsious cycle where human instilled biases result in AI bias which inturn train future AI models that reflect the same bias.

These biases do not only apply to races and ethnicities, it also applies to gender. Most medical experiments and tests have been done on male patients, which means that there are existing societal biases between men and women in healthcare.



A prominent example of this is cardiology. Heart attacks are overwhelmingly misdiagnosed in women, due to heart conditions presenting differently between males and females. Therefore AI models that claim to predict the likelyhood of heart attacks in an individual within the next 5 years may be inaccurate when predicting for women.

AI Bias in Healthcare



This kind of selection bias is also present between age groups. Most studies done about sleep disorders were conducted using younger patients as test subjects, meaning that AI models that were trained on that data have a difficult time deciphering sleep disorders in older patients.

These biases are further concerning in areas where the lack of variety is harder to detect such as drug development. When trials of new drugs are done on a group that is mainly focused on one ethnic group, a certain age range or gender can cause the features of the drug such as effects, side effects and effectiveness can transfer into AI models trained on this data. Creating bias that is harder to detect in the early stages of development.





The lack of diversity in training data is only one flaw that results in biased AI models. Unconscious bias from medical professionals may make itself into research, judgement and evaluations. These biased standpoints can weave itself into AI models overtime, causing bias.

The identification of such biases and efforts to remove them are crucial in maintaining a safe and fair healthcare system. There must a continuous search for gaps in the data, and investigations on which systems are biased. Research must be done to fill in these gaps and eliminate the systematic biases that AI models may have.



Case Study- Skin Lesions

A skin lesion is any part of the skin that differs from the tissue surrounding it. This could be a bump, redness, rashes or spots. Specific types of skin lesions are indications of more serious issues such as skin cancer, which means effective and quick identification of what these skin lesions are is extremely important.



Skin lesion identification softwares are trained on mostly white skin, with only 5~10% of training subjects being black. Therefore, when it comes to identifying skin lesions of black individuals, the accuracy rate decreases 50% compared to when identifying skin lesions on white skin. This issue is even more concerning considering that black people have a higher fatality rate due to melanoma, a dangerous type of skin cancer, compared to white patient. There is a 70% 5 year survival rate for black patients, which is significantly lower than the 90% of white individuals.

Ethical Data Harvesting

What is Data Harvesting?

Data harvesting is a process where companies collect data that is then used to train Artificial Intelligence models. This can be done through various methods, such as collecting data from social media posts that are under the jurisdiction of the company.

As Artificial Intelligence models require more and more data, companies are finding more and more ways to harvest data, bringing up questions about what the ideal, ethical practice is for such processes.

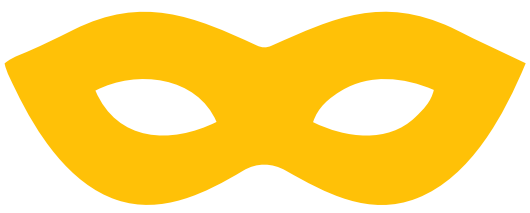
Issues with Data Harvesting

The Issue of Consent

Recieving explicit consent from the individual who's data is being collected is one of the most important principles. However, companies often use underhanded techniques that ensure that users are unaware that they are giving consent. For example, some companies may make their terms of agreement and privacy policies too long for users to read properly, resulting in users agreeing to such terms of user without explicit knowledge of what is being agreed upon. In addition, when additional data is being collected, or if the data that was consented towards being collected previously will be used for a new purpose, consent must also be collected.



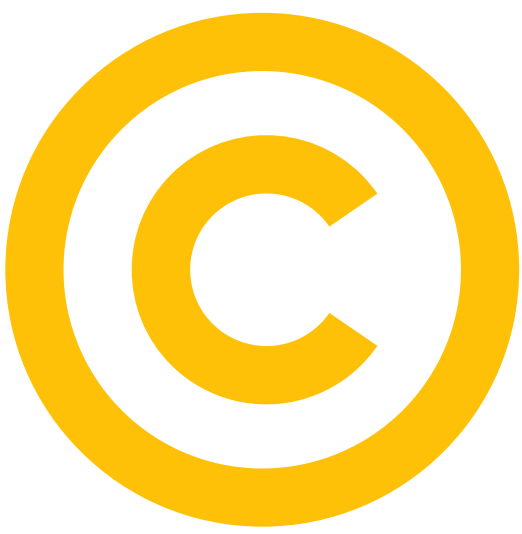
Privacy rights concerns - Anonymity



Another key principle in ethical data harvesting is to make sure that the data collected remains anonymous within the system. If the anonymity of the individual the data is collected from is not kept, there are risks of violating the right to privacy.

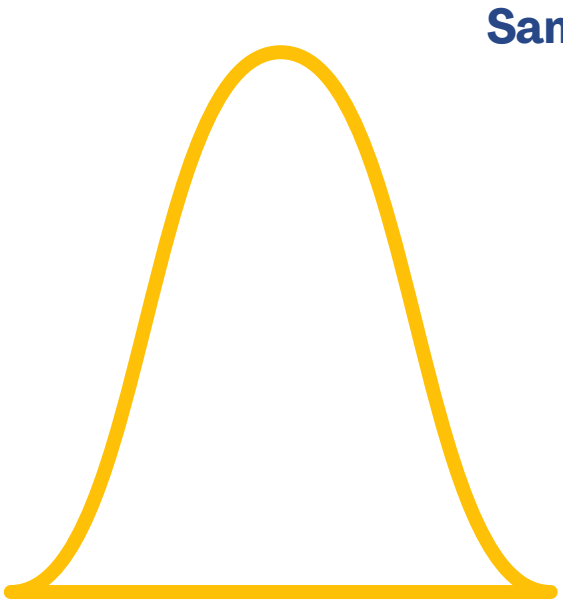
Copyright Concerns

Artists often use social media platforms to post art and promote their work. The harvesting of these images and using them to train data without copyright permission can cause legal issues of who owns images generated by these Gen-AI models.



An example of this is the case between Getty Images and Stability AI. Stability AI used copyrighted images of Getty Images to train their AI model. These concerns were dismissed, as the British court decided that Stability AI was using the copyrighted material as "transformative" purposes rather than "reproduction" purposes.

Sampling and Bias



When selecting data to use to train Artificial Intelligence, it is essential to sample data in order for it to be non-biased. Businesses and developers should be actively seeking out balanced data sets that can give fair insights to the AI model being trained.